Markov Chain

$X \to Y \to Z$, $Z \to Y \to X$

$p(z|y,x) = p(z|y)$, conditional indp $I(X;Z|Y) = 0$

C-K eq $p(z|x) = \sum_y p(z|y) \cdot p(y|x)$

Transition Matrix $P = [P_{ij}]$ $\pi^{(n)} = \pi^{(0)} P$ Homogenous : P_{ij} 与 t 无关

Steady state $\pi = \pi P$, $\sum \pi_j = 1$ $\Rightarrow$ π^*

$P^\infty = \pi^*_{1\times N} \otimes 1_{N\times 1} = \begin{bmatrix} \pi^* \\ \vdots \\ \pi^* \end{bmatrix}$

e.g. $P = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}$ $\circlearrowleft A \underset{\beta}{\overset{\alpha}{\rightleftarrows}} B \circlearrowright$

$\pi = \left(\frac{\beta}{\alpha+\beta} , \frac{\alpha}{\alpha+\beta} \right)$

Irreducible (communicate)

	Recurrent $\sum_{n=1}^{\infty} f_{jj}^{(n)} = 1$		Transient
	Null $M_j = \infty$	Non-null	
Periodic			
Aperiodic		Ergodic	

Finite state

$f_{jj}^{(n)} = P(n$ 次首达 $j)$

Recurrent $\sum_{n=1}^{\infty} f_{jj}^{(n)} = 1$

$M_j = \sum_{n=1}^{\infty} n f_{jj}^{(n)}$ 平均回来时间

Recurrent Null : $M_j = \infty$

RNN (Recurrent Non-Null) $M_j < \infty$

Ergodic 可遍历 Non-null + Aperiodic

Irreducible + Finite state $\Rightarrow$ Recurrent Non-null

e.g. $① \underset{1/2}{\overset{1}{\rightleftarrows}} ② \underset{1}{\overset{1/2}{\rightleftarrows}} ③$ RNN , not Ergodic

Irreducible

Period = 2

$f_{11}^{(n)} = \{0, \frac{1}{2}, 0, \frac{1}{4}, \cdots\}$

$\sum_{n=1}^{\infty} f_{11}^{(n)} = \sum_{n=1}^{\infty} \frac{1}{2^n} = 1$ Recurrent

$M_1 = \sum_{n=1}^{\infty} n f_{11}^{(n)} = 2\times\frac{1}{2} + 4\times\frac{1}{4} + 6\times\frac{1}{8} + \cdots = \sum_{n=1}^{\infty} \frac{2n}{2^n} = \frac{1}{(1-\frac{1}{2})^2} = 4 < \infty$

$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ $\sum_{n=1}^{\infty} x^n = \frac{x}{1-x}$

$\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}$

BDC (Birth Death Chain)

$$\circlearrowleft 0 \underset{\beta}{\overset{\alpha}{\rightleftarrows}} \overset{\circlearrowright}{1} \underset{\beta}{\overset{\alpha}{\rightleftarrows}} \overset{\circlearrowright}{2} \cdots$$

$\rho = \frac{\alpha}{\beta} > 1$ Transient, 发散

$\rho = 1$ Recurrent Null

$\rho < 1$ Recurrent Non-Null

$$\pi_j = \rho^j (1-\rho)$$

$\pi_0 = (1-\alpha)\pi_0 + \beta\pi_1$, $\pi_1 = \rho\pi_0$

$\pi_1 = \alpha\pi_0 + (1-\alpha-\beta)\pi_1 + \beta\pi_2$, $\pi_2 = \rho^2\pi_0$ ……

$\sum \pi_j = \pi_0(1+\rho+\rho^2+\cdots) = \frac{1}{1-\rho}\pi_0 = 1 \Rightarrow \pi_0 = 1-\rho$

e.g. $\circlearrowleft 0 \underset{\beta}{\overset{\alpha}{\rightleftarrows}} \overset{\circlearrowright}{1} \cdots \underset{\beta}{\overset{\alpha}{\rightleftarrows}} N \circlearrowright$ 求 π_j

$\pi_j = \rho^j \pi_0$ $\quad \sum \pi_j = \pi_0(1+\rho+\cdots+\rho^N) = \pi_0 \frac{1-\rho^{N+1}}{1-\rho} = 1$

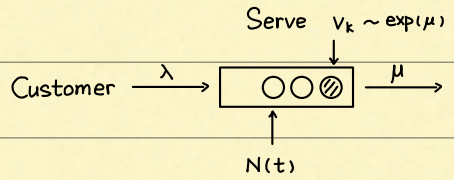
$\pi_0 = \frac{1-\rho}{1-\rho^{N+1}}$ $\quad \pi_j = \rho^j \frac{1-\rho}{1-\rho^{N+1}}$

Queue M/M/1

$P(\text{队列长度 } j) = \pi_j = \rho^j(1-\rho)$

平均等待时间 $\bar{W} = \frac{1}{\beta-\alpha}$

平均排队长度 $\bar{L} = \frac{\alpha}{\beta-\alpha} = \frac{\rho}{1-\rho}$



Entropy Rate $H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} H(X_1, \cdots, X_n)$

稳定 RP 的熵率必存在

Pf: $H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n-1} H(X_{i+1} | X_i, \cdots, X_1)$

$H(X_{i+1} | X_i, \cdots, X_1) \leq H(X_{i+1} | X_i, \cdots, X_2) = H(X_i | X_{i-1}, \cdots, X_1)$ SSS

因 $H(X_i | X_{i-1}, \cdots, X_1) \geq 0$，单调有界 $\lim_{i\to\infty} H(X_i | X_{i-1}, \cdots, X_1)$ 存在

若 $a_i \to a$，则 $\frac{1}{n} \sum_{i=1}^{n-1} a_i \to a$ 故 $H(\mathcal{X})$ 存在

时不变马尔科夫链 $H(\mathcal{X}) = H(X_2 | X_1) = \sum_i \mu_i H(i)$

Pf: $H(X_1, \cdots, X_n) = \sum_{i=1}^{n-1} H(X_{i+1} | X_i) \underset{\text{Markov}}{=} nH(X_2 | X_1)$

e.g. $P = \begin{vmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{vmatrix}$，求 $H(\mathcal{X})$

$\pi = \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta} \right)$

则 $H(\mathcal{X}) = H(X_2 | X_1) = \pi_1 H(\alpha) + \pi_2 H(\beta)$

$= \frac{\beta}{\alpha+\beta} H(\alpha) + \frac{\alpha}{\alpha+\beta} H(\beta)$